WORKED SOLUTION

STEP II 1992, Question 5

Pure • Matrix Groups

> **STEP II 1992, Question 5**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/92-s2-q5

The key idea

Everything here rests on the condition $wz - xy = 1$, which makes the inverse of a matrix in S especially simple. The order-3 part is best handled by noticing that every 2×2 matrix satisfies a quadratic equation built from its trace and determinant. For a matrix in S this lets us write $\mathbf{A}^2$, and then $\mathbf{A}^3$, in terms of $\mathbf{A}$ and $\mathbf{I}$, so the condition $\mathbf{A}^3 = \mathbf{I}$ becomes a condition on the trace $w + z$.

(The paper prints the bottom-right entry of $\mathbf{A}$ as x; the final part refers to z, so the intended matrix clearly has z there, and we use that throughout.)

Solution

The order of an element

The order of $g \in G$ is the smallest positive integer n with $g^n = e$, where e is the identity of G. If no positive power of g equals e, then g has infinite order. The identity is the only element of order 1.

The inverse

Let

$$\mathbf{A} = \begin{pmatrix} w & x \\ y & z \end{pmatrix}, \qquad wz - xy = 1.$$

The usual formula for a 2×2 inverse divides by the determinant, which here is 1, so

$$\mathbf{A}^{-1} = \begin{pmatrix} z & -x \\ -y & w \end{pmatrix}.$$

Multiplying out confirms this: $\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} wz - xy & 0 \\ 0 & wz - xy \end{pmatrix} = \mathbf{I}$, and similarly $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$.

S is a group

We check the four group axioms.

- *Closure.* If $\mathbf{A}, \mathbf{B} \in S$, then $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B} = 1 \times 1 = 1$, so $\mathbf{AB} \in S$.
- *Associativity.* We may assume this for matrix multiplication.
- *Identity.* $\det \mathbf{I} = 1$, so $\mathbf{I} \in S$, and $\mathbf{IA} = \mathbf{AI} = \mathbf{A}$ for every $\mathbf{A}$.
- *Inverses.* The matrix $\mathbf{A}^{-1}$ found above has determinant $wz - (-x)(-y) = wz - xy = 1$, so it lies in S.

Hence S is a group under matrix multiplication.

Self-inverse elements and elements of order 2

Comparing $\mathbf{A}^{-1}$ with $\mathbf{A}$ entry by entry, $\mathbf{A}^{-1} = \mathbf{A}$ exactly when

$$z = w, \qquad -x = x, \qquad -y = y,$$

that is, $x = y = 0$ and $z = w$. The determinant condition then reads $w^2 = 1$, so $w = \pm 1$. The self-inverse elements are therefore

$$\mathbf{A} = \mathbf{I} \quad \text{and} \quad \mathbf{A} = -\mathbf{I}.$$

An element has order 2 when $\mathbf{A}^2 = \mathbf{I}$ (equivalently $\mathbf{A}^{-1} = \mathbf{A}$) and $\mathbf{A} \neq \mathbf{I}$. So the only element of order 2 is $-\mathbf{I}$.

Elements of order 3

Write $t = w + z$ for the trace of $\mathbf{A}$. Squaring,

$$\mathbf{A}^2 = \begin{pmatrix} w^2 + xy & x(w+z) \\ y(w+z) & xy + z^2 \end{pmatrix}.$$

Since $xy = wz - 1$, the top-left entry is $w^2 + wz - 1 = tw - 1$, and the bottom-right is $tz - 1$. Hence

$$\mathbf{A}^2 = t\mathbf{A} - \mathbf{I}. \qquad (*)$$

Ask yourself How does $(*)$ help with $\mathbf{A}^3$?

Multiply $(*)$ by $\mathbf{A}$ and use $(*)$ again to remove the $\mathbf{A}^2$:

$$\mathbf{A}^3 = t\mathbf{A}^2 - \mathbf{A} = t(t\mathbf{A} - \mathbf{I}) - \mathbf{A} = (t^2 - 1)\mathbf{A} - t\mathbf{I}.$$

So any power of $\mathbf{A}$ is a combination of $\mathbf{A}$ and $\mathbf{I}$, and we can compare coefficients.

If $w + z + 1 = 0$: then $t = -1$, so $t^2 - 1 = 0$ and $\mathbf{A}^3 = \mathbf{I}$. Also $\mathbf{A} \neq \mathbf{I}$, because $\mathbf{I}$ has trace $2 \neq -1$. The order of $\mathbf{A}$ divides 3 and isn't 1, so it is 3.

Conversely, suppose $\mathbf{A}$ has order 3. Then $\mathbf{A}^3 = \mathbf{I}$ and $\mathbf{A} \neq \mathbf{I}$, so

$$(t^2 - 1)\mathbf{A} = (1 + t)\mathbf{I}.$$

- If $t^2 \neq 1$, then $\mathbf{A} = \dfrac{1+t}{t^2-1}\mathbf{I} = \dfrac{1}{t-1}\mathbf{I}$, a multiple $\lambda\mathbf{I}$ of the identity. Its determinant is $\lambda^2 = 1$, so $\lambda = \pm 1$. But $\mathbf{A} = \mathbf{I}$ is excluded, and $\mathbf{A} = -\mathbf{I}$ has order 2, not 3. So this case can't happen.
- If $t = 1$, the equation becomes $\mathbf{0} = 2\mathbf{I}$, which is false.

The only remaining possibility is $t = -1$, that is, $w + z + 1 = 0$. This proves the "if and only if".

An example. Take $w = 0$ and $z = -1$, and choose $x = -1$, $y = 1$ so that $wz - xy = 0 + 1 = 1$:

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad \mathbf{A}^2 = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \mathbf{A}^3 = \mathbf{I}.$$

Checks

- Multiply out the example above to see $\mathbf{A}^3 = \mathbf{I}$ directly.
- The rotation through 120°, $\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$, has determinant 1 and trace -1, and of course has order 3.
- We checked by computer that $\mathbf{A}^2 = t\mathbf{A} - \mathbf{I}$ for 20 000 random rational matrices in S. We also searched every matrix in S whose entries are multiples of $\frac{1}{2}$ between -4 and 4: exactly the 48 with trace -1 have order 3, and $-\mathbf{I}$ is the only one of order 2.

Common slips

- Including $\mathbf{I}$ among the elements of order 2. It is self-inverse, but its order is 1.
- Proving only one direction of the order-3 result. The converse needs the cases $t^2 \neq 1$ and $t = 1$ ruled out.
- Forgetting to check that the inverse itself has determinant 1 when verifying the group axioms.

Beyond STEP

The group S is the *special linear group* SL_2, one of the most important groups in mathematics. The identity $\mathbf{A}^2 - (\operatorname{tr}\mathbf{A})\mathbf{A} + (\det\mathbf{A})\mathbf{I} = \mathbf{0}$ is the 2×2 case of the Cayley–Hamilton theorem: every square matrix satisfies its own characteristic equation. It also shows how much the trace alone tells us about an element of SL_2: the argument above generalises to show that an element of finite order n has trace $2\cos(2\pi k/n)$ for some integer k. For more on groups of matrices, see Allenby, *Rings, Fields and Groups*; for Cayley–Hamilton and eigenvalues, see Strang, *Introduction to Linear Algebra*.

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